

A Nonlocal Poisson Boltzmann Ion Channel Model and Finite Element Solver

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Abstract

Voltage-Dependent Anion Channel (VDAC) is a class of channel proteins abundant on the outer mitochondrial membrane (OMM) playing a major role in moving crucial ions and metabolites into and out of mitochondria. We propose a nonlocal Poisson-Boltzmann ion channel (NPBIC) model to incorporate nonlocal dielectric effects into dielectric continuum modeling of VDACS. We also present a finite element method to solve the NPBIC model using tetrahedral meshes, x-ray crystallographic molecular structures of VDAC from the RCSB Protein Data Bank, and ionic solutions with multiple species.

Box Domain Partition and Meshes

We select a simulation box domain, Ω , containing a VDAC protein embedded in OMM and satisfying the partition

$$\Omega = D_p \cup D_m \cup D_s \cup \Gamma_p \cup \Gamma_m \cup \Gamma_{pm},$$

where D_p , D_s and D_m denote the protein, solvent, and membrane regions, respectively, Γ_p an interface between D_p and D_s ; Γ_m an interface between D_m and D_s ; and Γ_{pm} an interface between D_p and D_m . We generate an interface-fitted tetrahedral box domain mesh Ω_h , a protein region mesh $D_{p,h}$, a membrane region mesh $D_{m,h}$, and a solvent region mesh $D_{s,h}$ using our Ion Channel Mesh Package (ICMPv3) [1]. See Figure 1 for a mesh of Ω_h for the VDAC 3EMN.

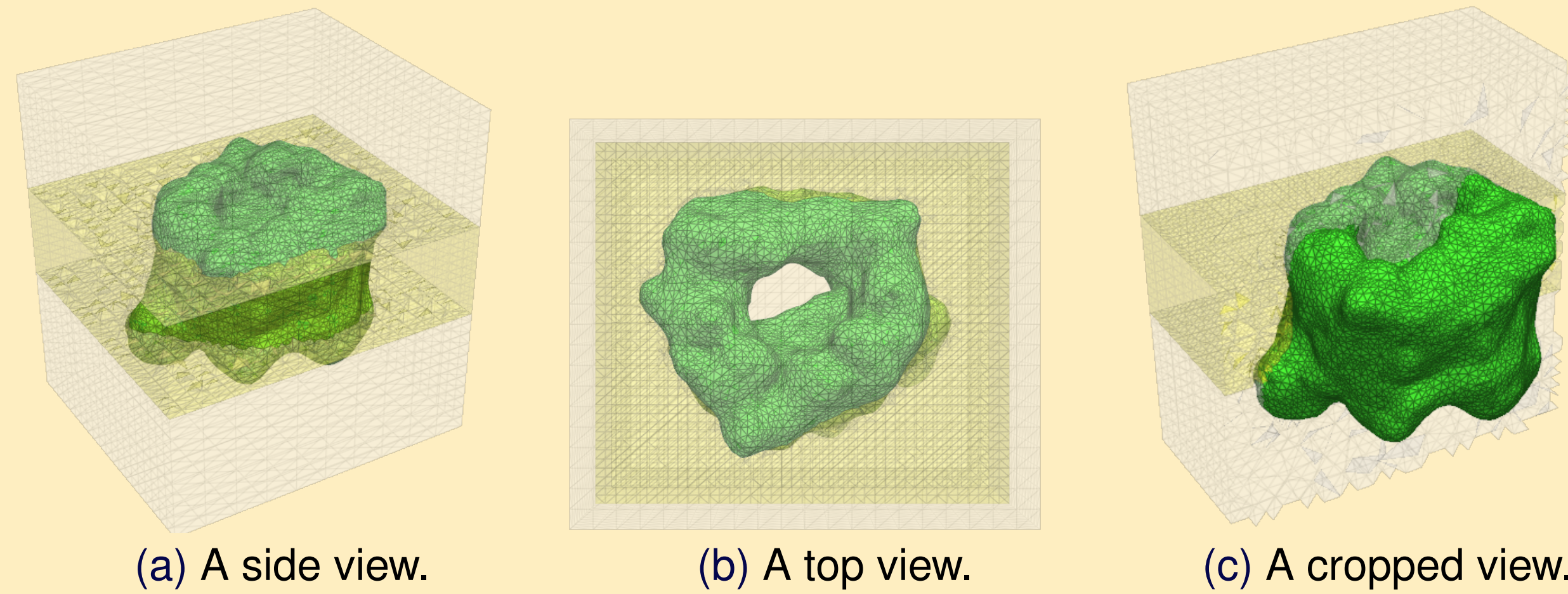


Figure 1: Three views of the box domain mesh Ω_h . Here, $D_{p,h}$, $D_{m,h}$, and $D_{s,h}$ are colored green, yellow, and gray, respectively, $D_{s,h}$ and $D_{m,h}$ are shown with low opacity to expose $D_{p,h}$ and $D_{m,h}$ partially; and $D_{s,h}$ is partially cropped to more clearly show the mesh structure of $D_{p,h}$.

Nonlocal Relationship of Displacement and Electric Fields

The local Poisson ion channel models are based on linear dielectric relationships $\mathbf{d}(\mathbf{r}) = \epsilon_0 \epsilon_p \mathbf{e}(\mathbf{r})$ in D_p , $\mathbf{d}(\mathbf{r}) = \epsilon_0 \epsilon_m \mathbf{e}(\mathbf{r})$ in D_m , and $\mathbf{d}(\mathbf{r}) = \epsilon_0 \epsilon_s \mathbf{e}(\mathbf{r})$ in D_s . Here, $\mathbf{d}$ is the displacement field, $\mathbf{e}$ is the electric field, and ϵ_0 , ϵ_p , ϵ_m , ϵ_s are permittivity constants. With Φ denoting the electrostatic potential function of $\mathbf{e}$, we obtain a nonlocal dielectric relationship within D_s as follows:

$$\mathbf{d}(\mathbf{r}) = \epsilon_0 \int_{\mathbb{R}^3} \epsilon(\mathbf{r} - \mathbf{r}') \nabla \Phi(\mathbf{r}') d\mathbf{r}', \quad \text{for } \mathbf{r} \text{ in } D_s$$

where ϵ is a nonlocal permittivity function given by

$$\epsilon(\mathbf{r}) = \epsilon_\infty \delta + (\epsilon_s - \epsilon_\infty) Q_\lambda(\mathbf{r}).$$

Here δ denotes the Dirac-delta distribution at the origin, ϵ_∞ and λ are nonlocal parameters, and $Q_\lambda(\mathbf{r}) = \frac{e^{-|\mathbf{r}|/\lambda}}{4\pi\lambda^2|\mathbf{r}|}$, which is a Yukawa-type kernel satisfying

$$-\lambda^2 \Delta Q_\lambda(\mathbf{r}) + Q_\lambda(\mathbf{r}) = \delta, \quad \mathbf{r} \in \mathbb{R}^3.$$

Nonlocal Poisson Boltzmann Ion Channel Model

We scale Φ as a dimensionless potential u to obtain the NPBIC model as follows:

$$\begin{aligned} -\epsilon_p \Delta u(\mathbf{r}) &= \alpha \sum_{j=1}^{n_p} z_j \delta_{r_j}(\mathbf{r}), & \mathbf{r} \in D_p, \\ -\epsilon_\infty \Delta u(\mathbf{r}) + \frac{\epsilon_s - \epsilon_\infty}{\lambda^2} [u(\mathbf{r}) - (u * Q_\lambda)(\mathbf{r})] &= \beta \sum_{i=1}^n Z_i c_i^b e^{-Z_i u(\mathbf{r})}, & \mathbf{r} \in D_s, \\ -\epsilon_m \Delta u(\mathbf{r}) &= 0, & \mathbf{r} \in D_m, \\ u(\mathbf{s}^-) &= u(\mathbf{s}^+), \quad \epsilon_p \frac{\partial u(\mathbf{s}^-)}{\partial \mathbf{n}_p(\mathbf{s})} = \epsilon_\infty \frac{\partial u(\mathbf{s}^+)}{\partial \mathbf{n}_p(\mathbf{s})} + (\epsilon_s - \epsilon_\infty) \frac{\partial (u * Q_\lambda)(\mathbf{s})}{\partial \mathbf{n}_p(\mathbf{s})}, & \mathbf{s} \in \Gamma_p, \\ u(\mathbf{s}^-) &= u(\mathbf{s}^+), \quad \epsilon_m \frac{\partial u(\mathbf{s}^-)}{\partial \mathbf{n}_m(\mathbf{s})} = \epsilon_\infty \frac{\partial u(\mathbf{s}^+)}{\partial \mathbf{n}_m(\mathbf{s})} + (\epsilon_s - \epsilon_\infty) \frac{\partial (u * Q_\lambda)(\mathbf{s})}{\partial \mathbf{n}_m(\mathbf{s})}, & \mathbf{s} \in \Gamma_m, \\ u(\mathbf{s}^-) &= u(\mathbf{s}^+), \quad \epsilon_p \frac{\partial u(\mathbf{s}^-)}{\partial \mathbf{n}_p(\mathbf{s})} = \epsilon_m \frac{\partial u(\mathbf{s}^+)}{\partial \mathbf{n}_p(\mathbf{s})}, & \mathbf{s} \in \Gamma_{pm}, \\ u(\mathbf{s}) &= g(\mathbf{s}), & \mathbf{s} \in \Gamma_D, \\ \frac{\partial u(\mathbf{s})}{\partial \mathbf{n}_b(\mathbf{s})} &= 0, & \mathbf{s} \in \Gamma_N, \end{aligned}$$

where α and β are constants, n_p is the number of atoms in a VDAC, $\mathbf{z}_j$ is the charge number of atom j at $\mathbf{r}_j$, n is the number of species of the ionic solution, Z_i is the charge number of species i , and $(\Phi * Q_\lambda)(\mathbf{r}) = \int_{\mathbb{R}^3} \Phi(\mathbf{r}') Q_\lambda(\mathbf{r} - \mathbf{r}') d\mathbf{r}'$.

Finite Element Solver and Program Package

We have developed an efficient NPBIC finite element solver and implemented it as a Python package, enhanced by Fortran extension modules, based on the finite element library of the FEniCS project (Version 2019.1.0), our SMPBic program package, and our nonlocal modified PBE program package. The NPBIC package adapts ICMPv3 [1] to produce meshes Ω_h , $D_{p,h}$, $D_{m,h}$ and $D_{s,h}$.

To avoid singularity difficulties caused by atomic charges, the NPBIC package calculate u by the solution decomposition formula

$$u = \mathbf{G} + \Psi + \tilde{\Phi},$$

where $\mathbf{G}$ collects all the singularity points of u as a known function, Ψ is the solution to a linear nonlocal interface boundary value problem and $\tilde{\Phi}$ is the solution to a nonlinear nonlocal interface boundary value problem. The function $\tilde{\Phi}$ is found by a modified Newton iterative method, using an initial guess $\tilde{\Phi}^{(0)}$ generated from a local PB ion channel solution.

Numerical Test Results

We conducted numerical experiments on two VDACS with PDB ID 3EMN and 2JK4. 3EMN is a mouse VDAC crystal structure, captured at 2.3 Å resolution, consisting of 2,250 atoms and 283 residues. 2JK4 is a human VDAC crystal structure, captured at 4.10 Å resolution, consisting of 2,207 atoms and 288 residues. For 3EMN, we set $\Omega = [-36, 43] \times [-36, 33] \times [-44, 40]$, and the mesh Ω_h contains 38,694 vertices. For 2JK4, the mesh Ω_h contains 38,583 vertices with $\Omega = [-39, 42] \times [-36, 37] \times [-50, 46]$. All tests were conducted in an ionic solution of 0.2 molar NaCl on one core of Mac mini with M4 chip. Test results are reported in Table 1 and Figures 1 to 4.

PDBID	Find $\mathbf{G}$, $\hat{\mathbf{G}}$, $\nabla \mathbf{G}$, $\nabla \hat{\mathbf{G}}$	Find Ψ	Find $\tilde{\Phi}^{(0)}$	Find $\tilde{\Phi}$	Total
3EMN	1.33	0.34	21.45	20.86	52.19
2JK4	1.34	0.35	14.21	13.60	44.07

Table 1: Performance of our NPBIC package for 3EMN and 2JK4 in terms of computer CPU time (in seconds). Here, "Total" denotes the total CPU time.

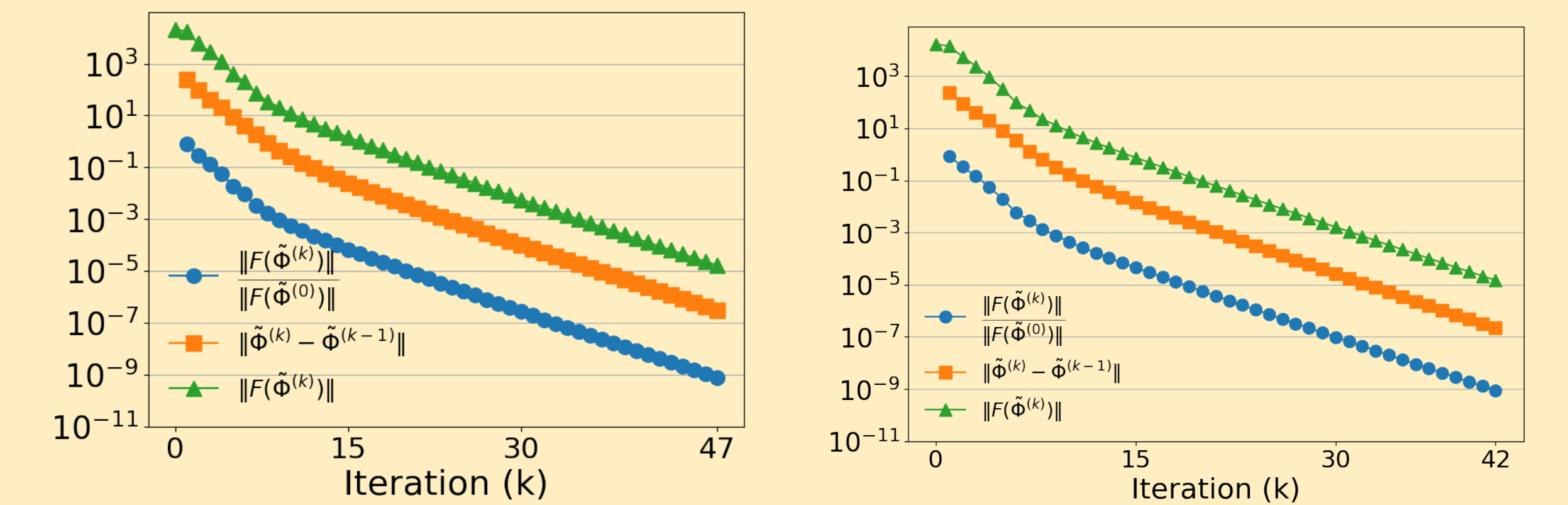


Figure 2: Convergence processes of our modified Newton iterative method for computing $\tilde{\Phi}$ in three errors: the relative residual error $\|F(\tilde{\Phi}^{(k)})\| / \|F(\tilde{\Phi}^{(0)})\|$, the error $\|\tilde{\Phi}^{(k)} - \tilde{\Phi}^{(k-1)}\|$, and the absolute residual error $\|F(\tilde{\Phi}^{(k)})\|$. Here $\tilde{\Phi}^{(k)}$ denotes the k th iterative approximation to $\tilde{\Phi}$.

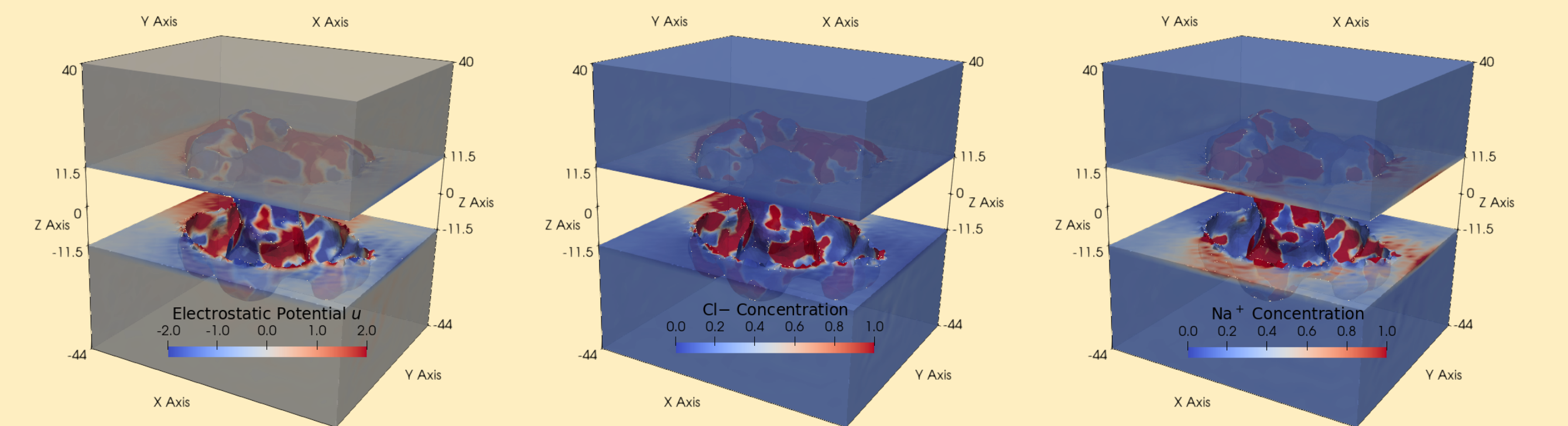


Figure 3: Color-mappings of the NPBIC finite element solution u and the concentrations of Na^+ and Cl^- in molar, determined by u , for 3EMN, confirming the anion selectivity of VDAC.

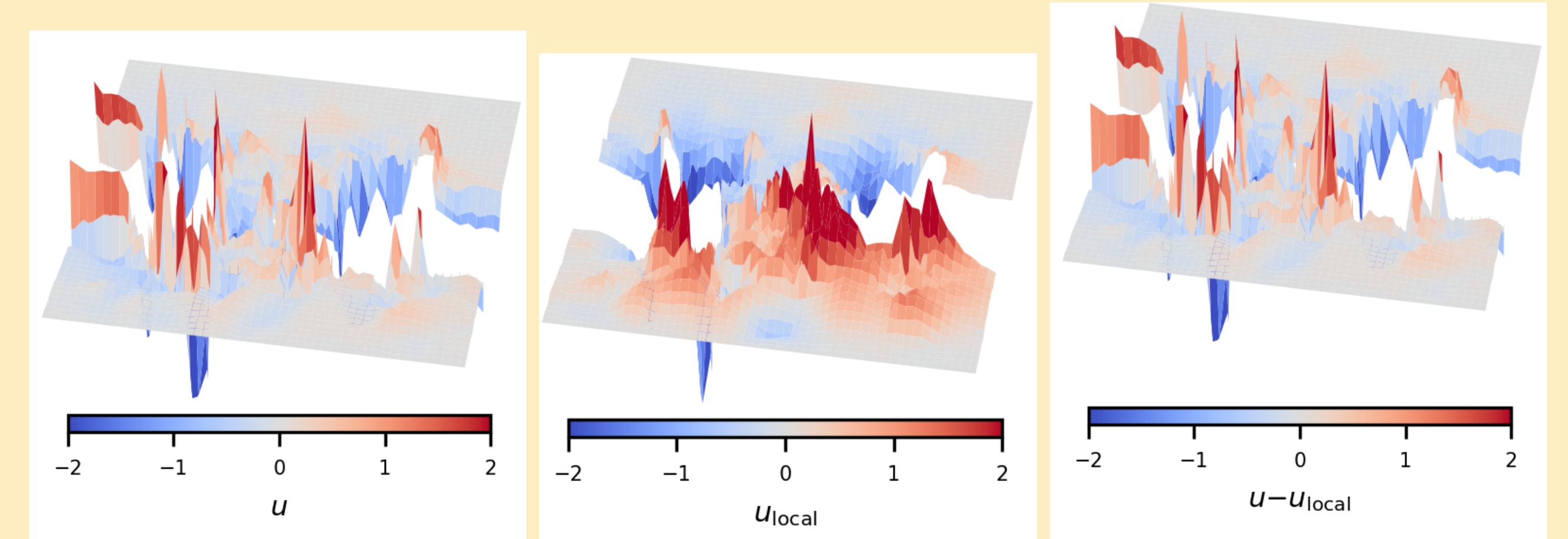


Figure 4: A comparison of the NPBic solution $u(0, y, z)$ with the local PBic solution $u_{loc}(0, y, z)$ for (y, z) in a slice $(x = 0)$ of D_s . Here u , u_{loc} and $u - u_{loc}$ are plotted as surfaces, showing that the NPBIC model produces a significantly different electrostatic potential function.

We plan to use our NPBIC package to compute electrostatic solvation free energies, making it a valuable tool in various biological applications.

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[1] L. Jemison, M. Stahl, R. K. Dash, and D. Xie, Vol. 41 (1), e70003, Journal of Computational Chemistry, 2025.